

Precalculus with Limits, 2e: Duolingo Math checklist

Tick each textbook section as you read it and each Duolingo unit as you finish it. Units are listed per lane (a Duolingo grade or topic) in the order the app presents them, so you can follow one lane top to bottom. Ratings: full = covers the section's main skill, partial = covers part of it, prereq = a warm-up from an earlier grade. The right column names the textbook sections the unit serves.

Chapter 1: Functions and Their Graphs (p. 1)

Grade 9 carries the basics (functions, lines, transformations), grade 11 repeats them at precalculus depth, and the Algebraic Graphing topic is a condensed walk through the whole chapter. Follow grade 11 if you only have time for one lane.

Textbook sections

☐	1.1 Rectangular Coordinates	p. 2
☐	1.2 Graphs of Equations	p. 13
☐	1.3 Linear Equations in Two Variables	p. 24
☐	1.4 Functions	p. 39
☐	1.5 Analyzing Graphs of Functions	p. 54
☐	1.6 A Library of Parent Functions	p. 66
☐	1.7 Transformations of Functions	p. 73
☐	1.8 Combinations of Functions: Composite Functions	p. 83
☐	1.9 Inverse Functions	p. 92
☐	1.10 Mathematical Modeling and Variation	p. 102

Book only, not on Duolingo: **1.1** Translating points in the plane; Applications of the distance and midpoint formulas. **1.2** Symmetry tests (x-axis, y-axis, origin); Sketching graphs of general equations by point plotting. **1.3** Slope as a rate of change in applications. **1.4** The difference quotient; Writing a function from a word problem. **1.5** Even and odd functions; The Vertical Line Test. **1.6** The square root and reciprocal parent functions in detail. **1.8** Domain of a composite function; Decomposing a function into two simpler functions. **1.9** One-to-one functions and the Horizontal Line Test. **1.10** Inverse, joint, and combined variation; Least squares regression on the calculator.

Grade 6

☐	33. Plot points in all quadrants	prereq	1.1
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Grade 7

☐	7. Proportionality in equations	prereq	1.10
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Grade 8

☐	21. Intro to slope	prereq	1.3
☐	22. Slope formula	prereq	1.3
☐	24. Slope-intercept form	prereq	1.3
☐	28. Equations of parallel lines	full	1.3
☐	29. Equations of perpendicular lines	full	1.3
☐	36. Intro to functions	prereq	1.4

Grade 9

☐	5. Slope-intercept form of linear equations	full	1.3
☐	7. Standard form of linear equations	full	1.3
☐	17. Inputs and outputs	full	1.4
☐	18. Graphs of functions	full	1.5
☐	20. Vertical intercept of functions	partial	1.2
☐	21. Horizontal intercept of functions	partial	1.2
☐	23. Average rate of change	full	1.5
☐	24. Linear functions	prereq	1.6
☐	25. Domain of a function	full	1.4
☐	26. Range of a function	partial	1.4
☐	27. Graphs of piecewise functions	full	1.5
☐	28. Equations of piecewise functions	full	1.4
☐	30. Step functions	full	1.6
☐	31. Absolute value functions	full	1.6
☐	33. Graphs of inverse functions	partial	1.9
☐	34. Equations of inverse functions	partial	1.9
☐	35. Graphs of inverse linear functions	prereq	1.9
☐	36. Equations of inverse linear functions	prereq	1.9
☐	75. Vertical translations	full	1.7
☐	76. Horizontal translations	full	1.7
☐	77. Vertical stretches	full	1.7
☐	78. Vertical compressions	full	1.7
☐	79. Horizontal stretches	full	1.7
☐	80. Horizontal compressions	full	1.7
☐	81. Scatter plots	partial	1.10
☐	82. Correlation	partial	1.10
☐	83. Linear models	partial	1.10
☐	84. Residuals	partial	1.10

Grade 10

☐	10. Reflectional symmetry	prereq	1.2
☐	45. Lines in point-slope form	full	1.3
☐	46. Lines in different forms	full	1.3
☐	47. Circles on the plane	full	1.2
☐	50. Line segment midpoint	full	1.1
☐	51. Distance formula	full	1.1

Grade 11

☐	2. Function families	full	1.6
☐	3. Compare function graphs	full	1.5
☐	9. Transform absolute value functions	full	1.7
☐	13. Relative maxima and minima of polynomial functions	partial	1.5
☐	65. Inverse function equations	full	1.9
☐	66. Inverse function graphs	full	1.9
☐	67. Combine functions	full	1.8
☐	68. Function composition	full	1.8
☐	69. Vertical translations	full	1.7
☐	70. Horizontal translations	full	1.7
☐	71. X-axis function reflections	full	1.7
☐	72. Y-axis function reflections	full	1.7
☐	73. Vertical stretches	full	1.7
☐	74. Vertical compressions	full	1.7
☐	75. Horizontal stretches	full	1.7
☐	76. Horizontal compressions	full	1.7

Grade 12

☐	13. Function composition	full	1.8
☐	14. Inverse functions	full	1.9
☐	17. Domain and range	full	1.5
☐	18. Graphing inverses	full	1.9

Geometry

☐	11. Absolute value and distance	prereq	1.1
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Algebraic Graphing

☐	3. Slope	partial	1.3
☐	4. Slope-intercept lines	partial	1.3
☐	8. Function notation	full	1.4
☐	9. Piecewise-defined functions	full	1.4
☐	10. Step and absolute value functions	full	1.6
☐	23. Inverse linear functions	partial	1.9
☐	24. Function translations	full	1.7
☐	25. Vertical function scaling	full	1.7
☐	26. Horizontal function scaling	full	1.7
☐	27. Reflections of functions	full	1.7

Statistics

☐	15. Linear models	partial	1.10
☐	16. Linear model fitness	partial	1.10

Chapter 2: Polynomial and Rational Functions (p. 125)

Grades 11 and 12 between them cover polynomial, complex, and rational functions thoroughly. Duolingo has no unit on polynomial long or synthetic division (2.3) or on nonlinear inequalities (2.7); use the book for those.

Textbook sections

☐	2.1 Quadratic Functions and Models	p. 126
☐	2.2 Polynomial Functions of Higher Degree	p. 136
☐	2.3 Polynomial and Synthetic Division	p. 150
☐	2.4 Complex Numbers	p. 159
☐	2.5 Zeros of Polynomial Functions	p. 166
☐	2.6 Rational Functions	p. 181
☐	2.7 Nonlinear Inequalities	p. 194

Book only, not on Duolingo: **2.1** Completing the square to write vertex form; Maximum and minimum word problems. **2.2** Using the Intermediate Value Theorem to locate a zero. **2.3** Polynomial long division; Synthetic division; The Remainder Theorem and the Factor Theorem. **2.4** Plotting complex numbers in the complex plane. **2.5** The Fundamental Theorem of Algebra; Descartes's Rule of Signs; Upper and lower bounds for real zeros. **2.6** Applications of rational functions. **2.7** Polynomial inequalities; Rational inequalities; The test-interval (sign chart) method.

Grade 9

☐	11. Solve one-variable inequalities	prereq	2.7
☐	12. Compound 'or' inequalities	prereq	2.7
☐	13. Compound 'and' inequalities	prereq	2.7
☐	56. Tables of quadratic functions	full	2.1
☐	57. Graphs of quadratic functions	full	2.1
☐	58. Equations of quadratic functions	full	2.1
☐	59. Zeros of quadratic functions	full	2.1
☐	60. Factored form	full	2.1
☐	61. Vertex	full	2.1
☐	62. Vertex form	full	2.1

Grade 11

	12. Intro to polynomial functions	full	2.2
	13. Relative maxima and minima of polynomial functions	full	2.2
	14. Zeros of polynomial functions	full	2.2, 2.5
	15. Multiplicity of polynomial functions	full	2.2
	16. End behavior of odd polynomial functions	full	2.2
	17. End behavior of even polynomial functions	full	2.2
	18. The imaginary number, i	full	2.4
	19. Square roots of negative numbers	full	2.4
	20. Patterns of exponentiating i	full	2.4
	21. Multiply complex numbers	full	2.4
	22. Complex roots	full	2.5
	23. Complex conjugates	full	2.4
	24. Intro to rational functions	full	2.6
	25. Y-intercept of rational functions	full	2.6
	26. X-intercepts of rational functions	full	2.6
	27. Vertical asymptotes of rational functions	full	2.6
	28. Rewrite rational functions	full	2.6
	29. End behavior approaching a horizontal line	full	2.6
	30. End behavior approaching $y=0$	full	2.6
	31. End behavior approaching slant asymptotes	full	2.6

Grade 12

	2. Polynomial function graphs	full	2.2
	3. Roots of polynomial functions	full	2.5
	4. Rational root theorem	full	2.5
	5. Polynomial function equations from graphs	full	2.2
	6. Vieta's formula for quadratic polynomials	partial	2.5
	7. Vieta's formula for cubic polynomials	partial	2.5
	8. Zeros of rational functions	full	2.6
	9. Vertical asymptotes of rational functions	full	2.6
	10. Horizontal asymptotes of rational functions	full	2.6
	11. Holes in rational functions	full	2.6
	12. Slant asymptotes of rational functions	full	2.6
	48. Intermediate value theorem	partial	2.2

Algebraic Graphing

☐	11. Intro to quadratic functions	full	2.1
☐	12. Quadratic graphs and equations	full	2.1
☐	13. Factored form of quadratics	full	2.1
☐	14. Vertex form of quadratics	full	2.1
☐	30. Polynomial function equations	full	2.2
☐	31. Polynomial function graphs	full	2.2
☐	32. Polynomial graphs and roots	full	2.2
☐	33. Rational function equations and graphs	full	2.6
☐	34. Rewrite and graph rational functions	full	2.6
☐	35. Rational function asymptotes	full	2.6
☐	36. Holes and slant asymptotes	full	2.6

Chapter 3: Exponential and Logarithmic Functions (p. 215)

Grade 11 covers exponentials and logarithms almost section by section. Duolingo stops at growth and decay, so the Gaussian, logistic, and logarithmic models of 3.5 come from the book only.

Textbook sections

☐	3.1 Exponential Functions and Their Graphs	p. 216
☐	3.2 Logarithmic Functions and Their Graphs	p. 227
☐	3.3 Properties of Logarithms	p. 237
☐	3.4 Exponential and Logarithmic Equations	p. 244
☐	3.5 Exponential and Logarithmic Models	p. 255

Book only, not on Duolingo: **3.1** Compound interest, n times per year and continuously. **3.4** Logarithmic equations; Checking for extraneous solutions. **3.5** Gaussian models; Logistic growth models; Logarithmic models (pH, Richter scale, decibels).

Grade 9

☐	70. Exponential function tables	prereq	3.1
☐	71. Exponential function graphs	prereq	3.1
☐	72. Exponential function equations	prereq	3.1

Grade 11

☐	38. Exponential function tables	full	3.1
☐	39. Exponential function graphs	full	3.1
☐	40. Exponential function equations	full	3.1
☐	41. Exponential growth	full	3.1, 3.5
☐	42. Exponential decay	full	3.1, 3.5
☐	48. Intro to the natural base	full	3.1
☐	49. Continuous growth models	full	3.1, 3.5
☐	50. Intro to logarithms	full	3.2
☐	51. Logarithm value	full	3.2
☐	52. Product rule	full	3.3
☐	53. Quotient rule	full	3.3
☐	54. Power rule	full	3.3
☐	55. Change of base formula	full	3.3
☐	56. Intro to logarithmic functions	full	3.2
☐	57. Logarithmic functions in base 10	full	3.2
☐	58. Logarithmic functions in base 2	full	3.2
☐	59. Logarithmic function graphs	full	3.2
☐	60. Logarithmic function intercepts	full	3.2
☐	61. The natural logarithm	full	3.2
☐	62. Natural logarithm graphs	full	3.2
☐	63. Solve exponential equations using logarithms	full	3.4
☐	64. Solve base e equations using logarithms	full	3.4

Grade 12

☐	15. Properties of logarithms	full	3.3
☐	16. Undo exponentials	partial	3.2, 3.4

Algebra

☐	18. Logarithms	full	3.2
☐	19. Product and quotient rules	full	3.3
☐	20. Properties of logarithms	full	3.3
☐	21. Solve logarithmic and exponential equations	full	3.4

Algebraic Graphing

☐	15. Intro to exponential functions	full	3.1
☐	16. Exponential graphs and equations	full	3.1
☐	18. Exponential growth and decay	partial	3.5
☐	19. The natural base, e	full	3.1
☐	20. Logarithmic functions	full	3.2
☐	21. Logarithmic function graphs	full	3.2
☐	22. Natural logarithmic function	full	3.2

Textbook: Ron Larson and Robert P. Hostetler. *Precalculus with Limits*, 2nd ed. (Pre-AP National Student Edition) Thomson Brooks/Cole, 2009. ISBN 978-0-538-73659-6. Duolingo inventory: 820 units, last verified 2026-10-07 on iOS. Only titles and unit names are reproduced, for reference; no textbook or Duolingo content is included. This checklist is licensed CC BY 4.0; see the repository NOTICE for third-party material.